

K2-30b

R-2504

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-30_191118 · created 2026-08-24T23:38:56+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458805.6506 +/- 0.0071 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.048
Transit depth (Rp/Rs)^2	2.33 +/- 1.46 [%]
Orbital Inclination (inc)	89.46 +/- 2.36
Airmass coefficient 1 (a1)	10.64 +/- 8.2
Airmass coefficient 2 (a2)	-0.98 +/- 0.54
Scatter in the residuals of the lightcurve fit is	86.42 %
Best Comparison Star	None
Optimal Aperture	2.76
Optimal Annulus	3.82
Transit Duration (day)	0.092 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.048 vs 0.1266 (deviation 0.55σ)
Duration fit vs published	0.092 d vs 0.0971 d (deviation 0.32σ)
Mid-time O–C	-13.5 min
Depth SNR	3.2
Residual scatter	86.42 %

Light curve

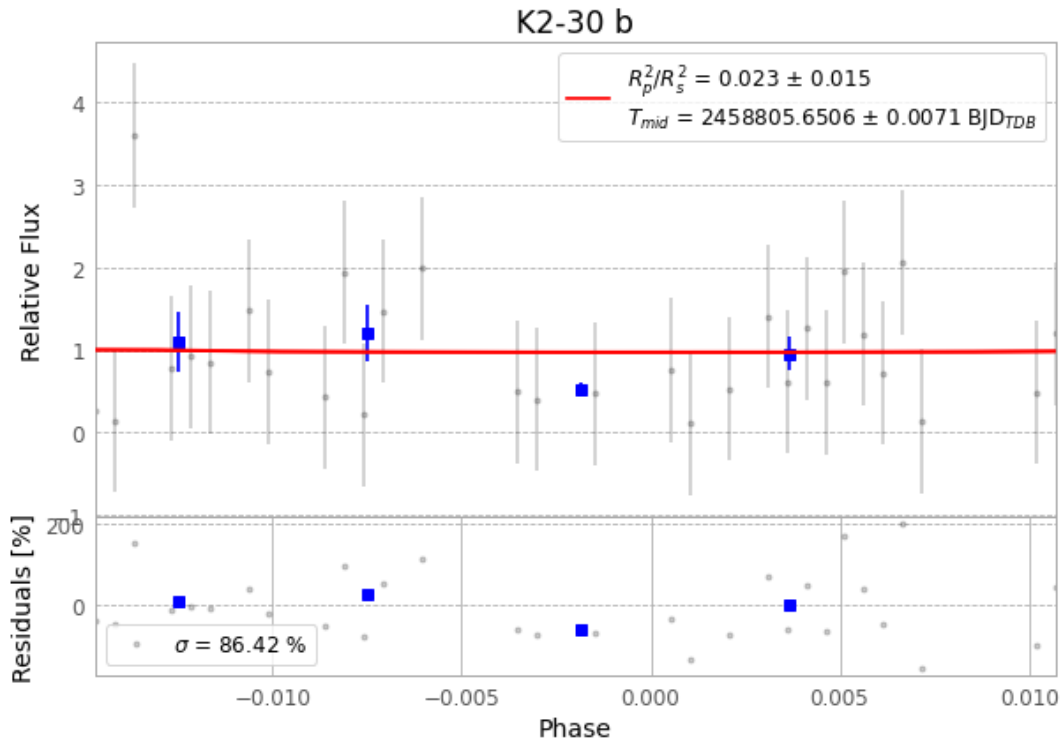


Figure 1 — FinalLightCurve_K2-30 b_2019-11-17.png (SHA-256 39aa3623e478b59a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [371, 201], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5c078c5acd13d1be...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

53 FITS frames (35 MB), K2-30191118015713.FITS → K2-30191118043312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-30-191118.log (6bad8f627492...), FinalLightCurve_K2-30 b_2019-11-17.png (39aa3623e478...), FinalLightCurve_K2-30 b_2019-11-17.csv (ed0445658d59...)

Compute resources

Wall time 112.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 23:38Z.